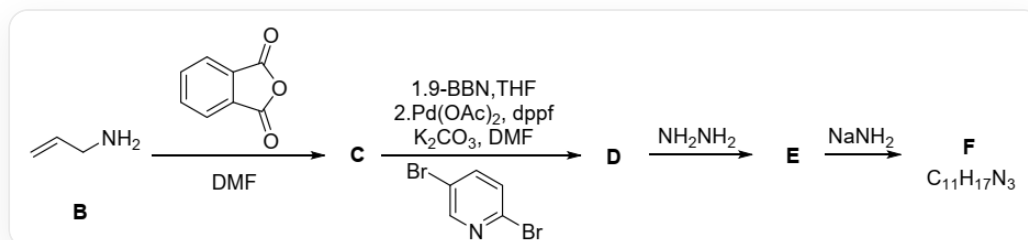


Question

Pyridine is more prone to aromatic nucleophilic substitution reactions than benzene. The following is the synthetic route of a certain important compound **X**:



The image describes an organic tandem reaction. The starting substrate **B** is $\text{C}=\text{CCN}$, which reacts with $\text{O}=\text{C1OC}(\text{C2}=\text{CC}=\text{CC}=\text{C21})=\text{O}$ and *DMF* to generate **C**. The reaction conditions for **C** to generate **D** are in 2 steps: 1. 9 – BBN; 2. $\text{Pd}(\text{OAc})_2$, *dppr*, K_2CO_3 , *DMF*, and $\text{BrC1}=\text{NC}=\text{C}(\text{Br})\text{C}=\text{C1}$. **D** reacts with NH_2NH_2 to generate **E**, and **E** reacts with NaNH_2 to generate **F**. The molecular formula of **F** is given in the figure as $\text{C}_{11}\text{H}_{17}\text{N}_3$.

The following statements are correct:

- A.** All other options are incorrect.
- B.** The reaction generating **E** is an aromatic nucleophilic substitution reaction.
- C.** **F** has three cycles
- D.** **F** exists a linkage relationship $\text{N} - \text{C} - \text{C} - \text{N}$
- E.** **F** has a six-membered ring and a five-membered ring.
- F.** **F** does not possess an independent amino group

Answer

Correct Answer: **A**

Detailed Explanation

The first step of the reaction is relatively simple, which is the Gabriel amidation reaction of the substrate, generating N-substituted phthalimide, the structure of **C** is O=C1N(CC=C)C(C2=CC=CC=C21)=O.

CHECKPOINT

1 PTS

The first step of the reaction is the Gabriel amidation of the substrate

CHECKPOINT

1 PTS

The structure of **C** is O=C1N(CC=C)C(C2=CC=CC=C21)=O

The second step is a typical Suzuki coupling reaction. First, **9 – BBN** is added to perform hydroboration on the allyl double bond, and then a *Pd(II)* catalyst is added to couple with the bromine of the substrate 2,5-dibromopyridine. Since the substrate has two bromines, a product in which both bromines are substituted may be obtained (this can also be seen from the chemical formula), and it is speculated that the structural formula of **D** is O=C1C2=C(C=CC=C2)C(N1CCCC3=NC=C(CCCN4C(C=C5C4=O)=O)C=C3)=O.

CHECKPOINT

1 PTS

The second step is a typical Suzuki coupling reaction

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1 PTS

The substrate has two bromines, so a product in which both bromines are substituted may be obtained

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1 PTS

It is speculated that the structural formula of **D** is
O=C1C2=C(C=CC=C2)C(N1CCCC3=NC=C(CCCN4C(C(C=CC=C5)=C5C4=O)=O)C=C3)=O

The reaction of **D** with N_2H_4 is the standard method for removing the phthalimide protecting group in Gabriel synthesis, producing an amine. Thus, the structure of **E** is NCCCC1=NC=C(CCCN)C=C1.

CHECKPOINT

1 PTS

The structure of **E** is NCCCC1=NC=C(CCCN)C=C1

When **E** is added to sodium amide, sodium amide as a strong base can abstract the proton on the amine group of the substrate; Since the 2/6 positions of pyridine are electron-deficient, the amino anion on the allyl substituent at the 5 position of pyridine can undergo aromatic nucleophilic substitution reaction, and the amino anion attacks the 6 position of pyridine to form a six-membered ring;

CHECKPOINT

1 PTS

The 2/6 positions of pyridine are electron-deficient, and the amino anion can undergo aromatic nucleophilic substitution reaction

However, after the formation of the six-membered ring, due to the addition of the substituent at the 6 position, the electron deficiency of pyridine is alleviated, and also because the 3 position is not the most electron-deficient position of pyridine; Therefore, the amino group of the substituent at the 2 position can no longer attack the 3 position to undergo a nucleophilic substitution reaction, so the product **F** only has two six-membered rings; the structure is NCCCC1=NC2=C(CCCN2)C=C1, which just matches the chemical formula.

CHECKPOINT

1 PTS

Due to the addition of the substituent at the 6 position, the electron deficiency of pyridine is alleviated, and also because the 3 position is not the most electron-deficient position of pyridine; Therefore, the amino group of the substituent at the 2 position can no longer attack the 3 position to undergo a nucleophilic substitution reaction

CHECKPOINT

1 PTS

F only has two six-membered rings; the structure is NCCCC1=NC2=C(CCCN2)C=C1

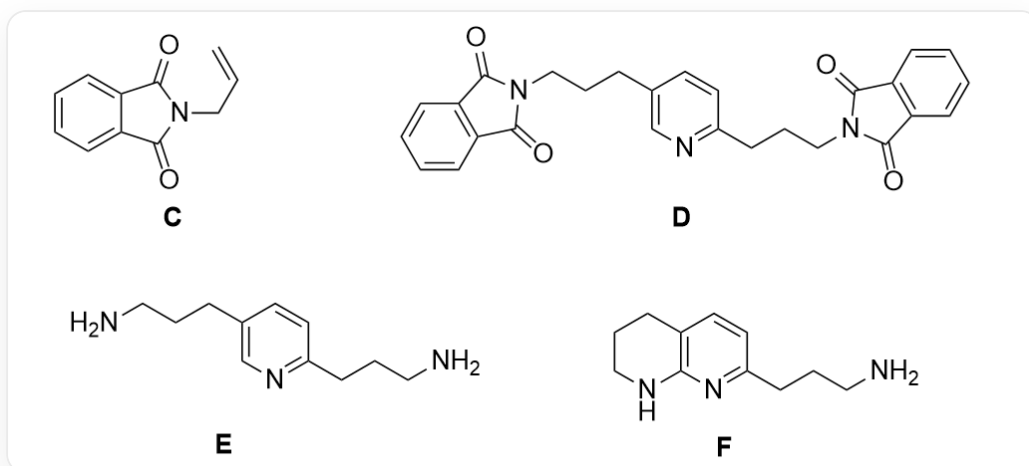
CHECKPOINT

1 PTS

F matches the chemical formula, the structure of **D** is speculated to be correct

Therefore, options B, C, E, and F are all incorrect; the structure of **F** does not have a N-C-C-N linkage, and option D is incorrect.

In summary, options B-F are all incorrect, and option A is correct.



本图片描述了本题涉及未知化合物的结构式，**C**结构为O=C1N(CC=C)C(C2=CC=CC=C21)=O，**D**的结构式为O=C1C2=C(C=CC=C2)C(N1CCCC3=NC=C(CCCN4C(C(C=CC=C5)=C5C4=O)=O)C=C3)=O，**E**结构为NCCCC1=NC=C(CCCN)C=C1，**F**结构为NCCCC1=NC2=C(CCCN2)C=C1